Troubleshooting Tools

Task 1 : Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

To identify the computer's IPv4 address and default gateway.

Command Used:

ipconfig

Network Details:

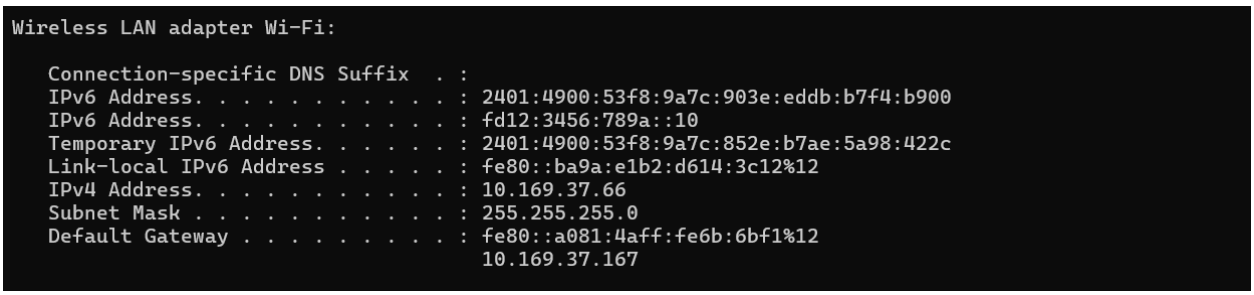
- IPv4 Address: 10.169.37.66
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.169.37.167

Explanation:

- **IPv4 Address:** Identifies my computer on the network.
- **Subnet Mask:** Determines the network and host portions of the IP address.
- **Default Gateway:** Allows my computer to communicate with other networks, such as the internet.

Conclusion: I successfully used the ipconfig command to find my computer's IPv4 address, subnet mask, and default gateway.

Screenshot:



Task 2 : Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

Command Used:

ping google.com

Procedure:

1. Open Command Prompt.
2. Type ping google.com and press Enter.
3. Wait for replies from Google's server.
4. Check the average response time in the ping statistics.

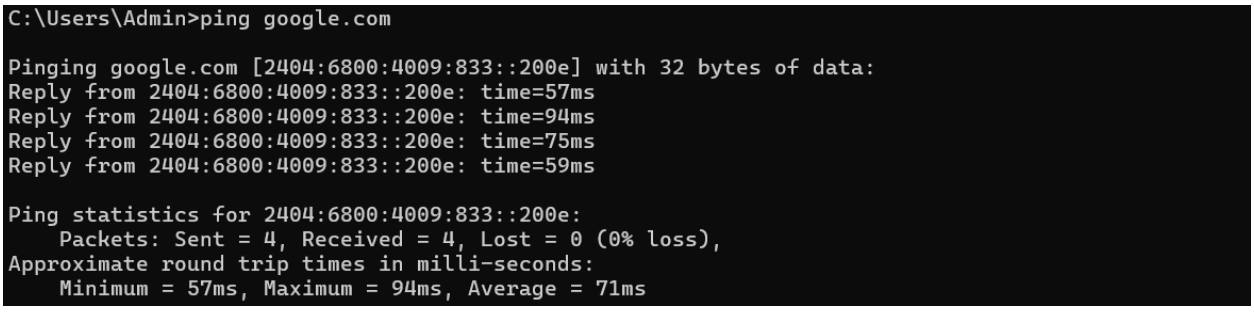
Result:

- Destination: google.com
- Connectivity: Successful
- Packets Sent: 4
- Packets Received: 4
- Packet Loss: 0%
- Minimum Response Time: 57 ms
- Maximum Response Time: 94 ms
- **Average Response Time: 71 ms**

Conclusion:

The ping command checks network connectivity between the computer and a destination server. It also measures the response time in milliseconds.

Screenshot:



Task 3 : Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

Command Used:

tracert www.flipkart.com

Procedure:

1. Open Command Prompt.
2. Type tracert www.flipkart.com and press Enter.
3. Wait for the command to complete.
4. Count the hops displayed in the output.

Result:

- Destination: www.flipkart.com
- Destination IP Address: 23.212.0.113
- Total Hops: 10
- Trace Status: Complete

Conclusion:

The tracert command shows the path taken by network packets to reach the destination. It identifies intermediate routers and helps troubleshoot network routing problems.

Screenshot:



Task 4 : Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

Command Used:

nslookup www.spotify.com

Procedure:

1. Open Command Prompt.
2. Type nslookup www.spotify.com and press Enter.
3. Observe the IP addresses displayed in the output.

Result:

- Domain Name: www.spotify.com
- DNS Server: dns.google
- DNS Server Address: 8.8.8.8
- IPv4 Address: 151.101.3.42
- Other IPv4 Addresses: 151.101.67.42, 151.101.131.42, 151.101.195.42
- DNS Resolution: Successful

Conclusion:

The nslookup command helps verify DNS resolution by checking whether a domain name is successfully translated into an IP address.

Screenshot:



Task 5 : A friend says they cannot access Zomato’s website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

If my friend cannot access Zomato's website from their laptop, I would use the following three commands to diagnose the issue.

1. ipconfig

ipconfig

Purpose: Checks the computer's IP address, subnet mask, and default gateway to verify network configuration.

2. ping

ping www.zomato.com

Purpose: Checks whether the computer can communicate with Zomato's server and measures the response time.

3. nslookup

nslookup www.zomato.com

Purpose: Checks whether the DNS server can resolve Zomato's domain name into an IP address.

Conclusion:

These three commands help identify network configuration problems, connectivity issues, and DNS resolution failures. They make it easier to troubleshoot website access problems.